PG CERTIFICATE PROGRAM IN

## AI / GenAI POWERED CYBERSECURITY

IIT ROORKEE × FUTURENSE

# CloudGuardian

## AI-Driven Cloud Misconfiguration Detection and Remediation

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Repository: https://github.com/MITRASENHUB/CLOUD_GUARDIAN_07

## Executive Summary

CloudGuardian is an AI-driven **Cloud Security Posture Management (CSPM)** solution developed as the capstone project for the **PG Certificate Program in AI/GenAI Powered Cybersecurity** offered by **IIT Roorkee in collaboration with Futurense**. The project demonstrates an end-to-end cloud security workflow that integrates Infrastructure as Code (IaC), cloud security assessment, machine learning, generative AI, and automated remediation into a unified framework for identifying, prioritizing, and mitigating cloud security misconfigurations.

Modern organizations increasingly rely on public cloud platforms such as **Microsoft Azure** and **Amazon Web Services (AWS)** to host critical workloads. While cloud platforms provide scalability, flexibility, and cost efficiency, they also introduce security challenges arising from configuration errors, excessive permissions, exposed storage resources, insecure networking rules, disabled monitoring, and inadequate encryption settings. Industry reports consistently identify cloud misconfiguration as one of the leading causes of data breaches. Traditional security assessments often generate hundreds of findings, making it difficult for security teams to identify which issues require immediate attention.

CloudGuardian addresses this challenge by implementing an intelligent security pipeline that not only detects cloud security weaknesses but also prioritizes them based on risk and recommends actionable remediation steps. The project combines rule-based cloud security scanning with artificial intelligence to reduce analyst effort and improve the speed and consistency of cloud security operations.

The project provisions cloud infrastructure using **Terraform**, ensuring that both secure and intentionally vulnerable environments can be deployed in a repeatable and reproducible manner. Separate environments were created in **Microsoft Azure** and **Amazon Web Services**, where controlled security misconfigurations were deliberately introduced to simulate real-world cloud security risks. These environments provide a realistic testbed for evaluating cloud security assessment tools and remediation workflows.

For security assessment, CloudGuardian utilizes multiple industry-recognized CSPM tools, including:

- **Prowler** for comprehensive cloud security benchmarking and compliance validation.
- **Steampipe** for SQL-based cloud configuration analysis and governance reporting.
- **ScoutSuite** for multi-cloud security auditing and visualization.

Using multiple scanning tools improves coverage and enables cross-validation of findings across different cloud services.

The Azure implementation includes **13 intentionally introduced security misconfigurations** spanning storage accounts, networking, identity management, SQL databases, encryption, and logging configurations. The AWS environment contains **14 security findings** involving Amazon S3, RDS, IAM, EC2 Security Groups, CloudTrail, and other foundational cloud services. These controlled vulnerabilities simulate common configuration errors encountered in enterprise cloud deployments.

Rather than treating all security findings equally, CloudGuardian incorporates a **Machine Learning–based prioritization engine** developed using the **Random Forest classification algorithm**. The model evaluates multiple characteristics of each finding—including severity, exploitability, exposure level, affected resource type, compliance impact, and potential blast radius—to predict remediation priority. This enables security teams to focus first on the vulnerabilities that present the greatest operational and business risk.

To improve usability and reduce manual analysis effort, the project integrates **Large Language Models (LLMs)** for automated security guidance. For every detected finding, the LLM generates concise, human-readable explanations describing:

- The security issue and why it is important.
- Potential business and operational impact.
- Recommended remediation actions.
- Best-practice implementation guidance.

This significantly reduces the time required for security analysts to interpret technical findings while making reports more accessible to infrastructure engineers and management stakeholders.

CloudGuardian further extends traditional CSPM capabilities by implementing a **controlled automated remediation framework**. Safe and low-risk security issues can be remediated automatically, while high-impact configuration changes are protected through **human approval gates** before execution. This human-in-the-loop design balances automation with governance, preventing unintended service disruptions while maintaining operational control.

To ensure regulatory alignment, every identified finding is mapped against internationally recognized security and privacy standards, including:

- **ISO/IEC 27001:2022** security controls.
- **Digital Personal Data Protection (DPDP) Act, 2023 (India)**.
- **HIPAA Security Rule** for healthcare information protection.
- **PCI-DSS** requirements for payment card security.

This compliance mapping enables organizations to understand not only the technical implications of security findings but also their potential regulatory and audit impact.

Overall, CloudGuardian demonstrates how **Infrastructure as Code, Cloud Security Posture Management, Machine Learning, Generative AI, and automated remediation** can be integrated into a single intelligent security workflow. The project provides a practical proof of concept for modern cloud security operations by enabling continuous detection, intelligent prioritization, explainable remediation guidance, governance through human approval, and compliance-aware reporting across multi-cloud environments. The resulting framework illustrates how AI can enhance the efficiency, consistency, and effectiveness of cloud security management while supporting secure and compliant cloud adoption.

## 1. Introduction & Problem Statement

### 1.1 Introduction

Cloud computing has transformed the way organizations develop, deploy, and manage applications by providing scalable, flexible, and cost-effective infrastructure. Leading cloud service providers such as **Microsoft Azure** and **Amazon Web Services (AWS)** enable businesses to rapidly provision resources and deliver services with minimal infrastructure management. However, this flexibility also introduces new security challenges. Cloud environments are highly dynamic, with resources being continuously created, modified, and removed through Infrastructure as Code (IaC), DevOps pipelines, and automated deployment processes.

Unlike traditional on-premises infrastructure, cloud environments can change within minutes. A single configuration error—such as a publicly accessible storage account, an overly permissive security group, disabled logging, weak identity permissions, or unencrypted databases—can expose sensitive data and create significant security vulnerabilities. As organizations scale their cloud adoption, maintaining a secure cloud posture becomes increasingly difficult.

Cloud Security Posture Management (CSPM) has emerged as a critical approach for continuously monitoring cloud environments, identifying security weaknesses, and ensuring compliance with industry standards. However, existing CSPM tools often generate large volumes of security findings without sufficient prioritization or contextual guidance. Security teams must manually analyze hundreds of alerts, determine which findings pose the greatest business risk, and identify the appropriate remediation steps. This process is time-consuming, resource-intensive, and susceptible to human error.

CloudGuardian was developed to address these challenges by combining **Infrastructure as Code (Terraform), Cloud Security Posture Management, Machine Learning, Generative AI, and controlled automation** into a single intelligent cloud security framework. Rather than simply detecting security issues, CloudGuardian prioritizes findings based on risk, explains them in natural language, recommends remediation actions, and automates safe fixes while maintaining governance through human approval for high-risk operations.

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## 1.2 Problem Statement

Cloud security posture degrades continuously as organizations expand their cloud infrastructure and deploy new services. Security configurations that are initially compliant can quickly become vulnerable due to frequent infrastructure changes, application updates, permission modifications, and evolving business requirements. Consequently, periodic manual security audits are no longer sufficient to identify and mitigate risks in modern cloud environments.

Several challenges contribute to this problem:

- Cloud resources are provisioned and modified rapidly, causing security configurations to drift from established best practices.
- Traditional security assessments provide only periodic snapshots and cannot keep pace with continuously changing cloud environments.
- CSPM tools often generate a large number of alerts without effectively distinguishing between critical and low-risk findings.
- Security analysts spend significant time manually reviewing reports, prioritizing vulnerabilities, and identifying suitable remediation actions.
- Many organizations hesitate to implement automated remediation because incorrect changes can disrupt production workloads.
- Compliance with standards such as **ISO/IEC 27001:2022**, the **Digital Personal Data Protection (DPDP) Act, 2023**, **HIPAA**, and **PCI-DSS** requires continuous monitoring and evidence-based reporting, which is difficult to maintain manually.

As cloud environments continue to grow in size and complexity, these challenges increase operational risk, delay vulnerability remediation, and raise the likelihood of security incidents, regulatory non-compliance, and data breaches.

CloudGuardian addresses these issues by providing a comprehensive AI-assisted cloud security workflow that continuously detects cloud misconfigurations, intelligently prioritizes findings using machine learning, generates explainable remediation guidance through Large Language Models (LLMs), automates low-risk corrective actions with appropriate approval mechanisms, and maps findings to recognized security and compliance frameworks.

### 1.3 Project Deliverables

The primary objective of CloudGuardian is to demonstrate a complete end-to-end Cloud Security Posture Management (CSPM) lifecycle across Microsoft Azure and Amazon Web Services. The major deliverables of the project include:

1. **Terraform-Based Infrastructure Provisioning**
    - Automated deployment of cloud infrastructure using Infrastructure as Code (IaC).
    - Creation of both secure and intentionally vulnerable cloud environments.
    - Repeatable and version-controlled infrastructure deployment.
2. **Cloud Security Posture Assessment**
    - Continuous security scanning using industry-standard CSPM tools including **Prowler**, **Steampipe**, and **ScoutSuite**.
    - Identification of cloud security misconfigurations across Azure and AWS resources.
    - Generation of detailed security assessment reports.
3. **Machine Learning–Based Risk Prioritization**
    - Development of a **Random Forest** model to prioritize detected vulnerabilities.
    - Risk scoring based on factors such as severity, exposure, exploitability, affected resource type, and potential business impact.
    - Intelligent ranking of findings to support efficient remediation planning.
4. **LLM-Based Security Guidance**
    - Automatic generation of human-readable explanations for each security finding.
    - Context-aware remediation recommendations.
    - Simplified technical reporting suitable for both security professionals and management.
5. **Controlled Automated Remediation**
    - Automation of predefined low-risk remediation actions.
    - Human approval gates for high-impact configuration changes.
    - Secure and governed implementation of corrective actions.
6. **Compliance Mapping**
    - Mapping identified findings to **ISO/IEC 27001:2022**, **DPDP Act 2023**, **HIPAA**, and **PCI-DSS** control requirements.
    - Generation of compliance-oriented security reports to support governance, auditing, and regulatory readiness.

Through these deliverables, CloudGuardian demonstrates a practical framework for modern cloud security management that integrates continuous monitoring, intelligent risk assessment, AI-assisted decision support, automated remediation, and compliance management into a unified solution for multi-cloud environments.

## 2. Architecture & Infrastructure

### 2.1 System Architecture

CloudGuardian is an AI-driven Cloud Security Posture Management (CSPM) framework that continuously detects cloud misconfigurations, prioritizes risks using Machine Learning, generates remediation guidance with Large Language Models (LLMs), and performs controlled automated remediation. The framework integrates Infrastructure as Code (Terraform), CSPM tools, AI, and compliance mapping to provide an end-to-end cloud security workflow across Microsoft Azure and Amazon Web Services.

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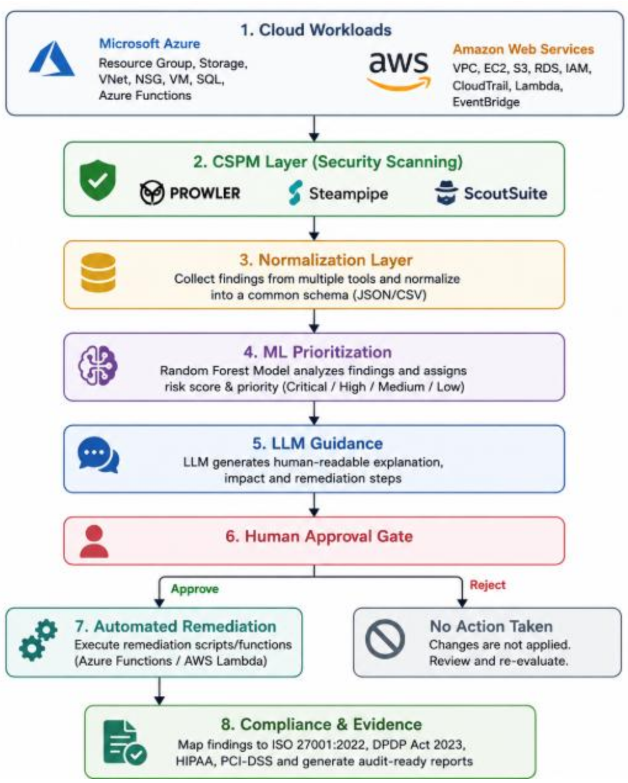
### 2.2 Overall Architecture Flow




### 2.3 Cloud Infrastructure

**Environment 1 – Microsoft Azure**

The Azure environment was provisioned using Terraform and includes a Resource Group, Virtual Network (VNet), Network Security Group (NSG), Virtual Machine (VM), Storage Account, Azure SQL Database, and Azure Functions. Thirteen intentional misconfigurations were introduced for security assessment.

**Environment 2 – Amazon Web Services (AWS)**

The AWS environment includes a VPC, EC2 instance, Amazon S3, RDS, IAM, CloudTrail, AWS Lambda, and EventBridge. Fourteen security findings were intentionally created to evaluate CSPM detection and remediation.

### 3.4 Technology Stack

| Layer | Technologies |
|---|---|
| Cloud Platforms | Microsoft Azure, AWS |
| Infrastructure | Terraform |
| Programming | Python |
| Data Analysis | Pandas |
| Machine Learning | Scikit-Learn (Random Forest) |
| CSPM Tools | Prowler, Steampipe, ScoutSuite |
| AI | OpenAI LLM |
| Automation | Azure Functions, AWS Lambda |
| Compliance | ISO 27001:2022, DPDP Act 2023, HIPAA, PCI-DSS |

## 3. Methodology & Codebase Structure

### 3.1 Methodology

CloudGuardian was developed using a phased approach over three weeks, following the complete Cloud Security Posture Management (CSPM) lifecycle. The methodology begins with infrastructure provisioning, followed by security assessment, intelligent risk prioritization, automated remediation, and compliance reporting.

- **Week 1 – Infrastructure Deployment & Misconfiguration Setup:**
  Cloud infrastructure was provisioned on Microsoft Azure and Amazon Web Services using Terraform. Secure resources were deployed, followed by the introduction of controlled misconfigurations to simulate real-world security vulnerabilities.
- **Week 2 – Security Assessment & Risk Prioritization:**
  CSPM tools including **Prowler**, **Steampipe**, and **ScoutSuite** were used to scan both cloud environments. The detected findings were normalized and

processed using a **Random Forest** machine learning model to prioritize vulnerabilities based on risk.

- **Week 3 – Remediation & Compliance:**
  Large Language Models (LLMs) generated remediation guidance for each finding. Automated remediation scripts with human approval gates were implemented, and all findings were mapped to **ISO/IEC 27001:2022**, **DPDP Act 2023**, **HIPAA**, and **PCI-DSS** to generate compliance reports.

### 3.2 Codebase Structure

The project repository is organized into modular directories, each responsible for a specific component of the CloudGuardian framework. This modular design improves maintainability, scalability, and ease of deployment.

| Directory | Purpose |
|---|---|
| **Azure/** | Terraform templates and deployment scripts for Microsoft Azure |
| **AWS/** | Terraform templates and infrastructure for Amazon Web Services |
| **Scans/** | CSPM scan reports generated using Prowler, Steampipe, and ScoutSuite |
| **ML/** | Machine Learning models, datasets, and prioritization scripts |
| **LLM/** | AI-powered remediation guidance and explanation generation |
| **Compliance/** | Mapping of findings to ISO 27001, DPDP, HIPAA, and PCI-DSS |
| **Evidence/** | Screenshots, reports, logs, and supporting project documentation |

This structured repository enables each phase of the project—from infrastructure deployment to compliance reporting—to be managed independently while supporting an integrated end-to-end cloud security workflow.

## 4. Cross-Environment Performance & Results

CloudGuardian was evaluated across both Microsoft Azure and Amazon Web Services (AWS) to validate its ability to detect, prioritize, and remediate cloud security misconfigurations in a multi-cloud environment. The security assessments were performed using Prowler, Steampipe, and ScoutSuite, which successfully identified configuration weaknesses across storage, networking, identity, databases, encryption, and logging services.

The Azure environment contained 13 intentionally introduced misconfigurations, while the AWS environment produced 14 security findings. These controlled vulnerabilities simulated common cloud security issues encountered in production environments and demonstrated the effectiveness of the CSPM tools in detecting configuration drift and security risks.

Representative findings identified during the assessment include:

- Publicly accessible Amazon S3 buckets
- Publicly accessible Amazon RDS instances
- Open SSH access (Port 22) in Security Groups and Network Security Groups
- Overly permissive IAM policies with wildcard (*) permissions
- Storage accounts and databases without encryption enabled
- Public network access enabled for cloud resources
- Disabled logging and monitoring configurations
- Weak identity and access management settings

Following detection, all findings were normalized and processed by the Random Forest Machine Learning model, which prioritized vulnerabilities based on multiple security attributes, including:

- Severity level
- Public exposure
- Exploitability
- Potential blast radius
- Resource type
- Compliance impact

The prioritization model classified findings into Critical, High, Medium, and Low priority levels, enabling security teams to focus remediation efforts on the most impactful risks first. This intelligent ranking reduced manual effort, improved remediation efficiency, and provided a structured approach for managing security findings across both Azure and AWS environments.

The evaluation demonstrates that CloudGuardian successfully integrates multi-cloud security assessment, AI-driven risk prioritization, and compliance-aware reporting into a unified workflow, providing a practical solution for modern cloud security operations.

## 5. Engineering Challenges & Student Effort

The development of CloudGuardian involved integrating multiple cloud platforms, security assessment tools, machine learning models, and automation workflows. Several technical challenges were encountered during implementation, requiring extensive troubleshooting and iterative testing.

Some of the key challenges included:

- **Python and Prowler Compatibility:** Managing Python virtual environments, package dependencies, and compatibility issues while installing and running the latest version of Prowler.

- **Terraform State Management:** Resolving Terraform state conflicts, resource dependencies, and deployment errors during repeated infrastructure provisioning and updates.

- **AWS Configuration:** Configuring AWS credentials, IAM permissions, and CLI authentication to enable secure deployment and CSPM scanning.

- **Azure Authentication:** Setting up Azure CLI authentication, managing subscriptions, and configuring access permissions for Terraform deployments and security assessments.

- **Lambda and Azure Function Deployment:** Packaging Python dependencies, handling runtime configurations, and deploying serverless functions for automated remediation.

- **CSPM Data Normalization:** Consolidating findings from multiple CSPM tools (Prowler, Steampipe, and ScoutSuite) into a standardized format suitable for machine learning and reporting.

- **Multi-Cloud Integration:** Maintaining consistent workflows across Azure and AWS despite differences in resource structures, security controls, and APIs.

- **Compliance Mapping:** Mapping security findings to multiple regulatory frameworks, including **ISO/IEC 27001:2022**, **DPDP Act 2023**, **HIPAA**, and **PCI-DSS**, while ensuring traceability and reporting consistency.

These challenges provided practical experience in cloud infrastructure management, cybersecurity operations, automation, and AI integration, significantly enhancing the robustness of the final solution.

Estimated team effort exceeded 70 hours across implementation, testing, remediation, compliance mapping, and documentation.

Major activities included:

- Infrastructure provisioning using Terraform on Azure and AWS

- Introducing and validating controlled cloud misconfigurations

- Executing CSPM security scans using Prowler, Steampipe, and ScoutSuite
- Developing and testing the Machine Learning prioritization model
- Integrating LLM-based remediation guidance
- Implementing automated remediation workflows with human approval gates
- Performing compliance mapping and audit evidence collection
- Preparing technical documentation, reports, screenshots, and project demonstrations

## 6. Conclusion

CloudGuardian successfully demonstrates an end-to-end AI-driven Cloud Security Posture Management (CSPM) framework capable of operating across both Microsoft Azure and Amazon Web Services (AWS). By integrating Infrastructure as Code (Terraform), CSPM security assessment tools, Machine Learning, Large Language Models (LLMs), and automated remediation, the project provides a practical approach to continuously identifying, prioritizing, and mitigating cloud security misconfigurations.

The project validates that a unified security workflow can be applied consistently across multiple cloud providers while maintaining governance through human approval gates and generating audit-ready compliance evidence. The use of Machine Learning improves the prioritization of security findings, enabling organizations to focus remediation efforts on the most critical risks. Additionally, LLM-generated remediation guidance simplifies the interpretation of technical findings and accelerates the remediation process.

CloudGuardian also demonstrates alignment with major security and regulatory frameworks, including ISO/IEC 27001:2022, DPDP Act 2023, HIPAA, and PCI-DSS, making the framework suitable for compliance-focused cloud environments.

Overall, the project highlights how AI can enhance modern cloud security operations by combining continuous monitoring, intelligent risk assessment, automated remediation, and compliance reporting into a scalable and portable multi-cloud solution. It serves as a strong proof of concept for next-generation cloud security platforms and provides a solid foundation for future enhancements such as real-time monitoring, predictive risk analytics, and support for additional cloud providers.

## 7. Appendix & Setup Guide

### Deployment Commands

| Step | Command | Purpose |
|---|---|---|
| 1 | az login | Authenticate with Azure |
| 2 | terraform init | Initialize Terraform providers and modules |
| 3 | terraform validate | Validate Terraform configuration *(Optional)* |
| 4 | terraform plan | Preview infrastructure changes |
| 5 | terraform apply | Deploy Azure infrastructure |
| 6 | prowler azure | Execute Azure CSPM security scan |

### Amazon Web Services (AWS) Deployment

| Step | Command | Purpose |
|---|---|---|
| 1 | aws configure | Configure AWS credentials |
| 2 | terraform init | Initialize Terraform providers |
| 3 | terraform validate | Validate Terraform configuration *(Optional)* |
| 4 | terraform plan | Preview infrastructure changes |
| 5 | terraform apply | Deploy AWS infrastructure |
| 6 | prowler aws | Execute AWS CSPM security scan |

### Standards & References

| Category | Reference |
|---|---|
| Cyber Threat Framework | MITRE ATT&CK® Framework |
| Cybersecurity Framework | NIST Cybersecurity Framework (CSF) 2.0 |
| Cloud Security Benchmarks | CIS Benchmarks |
| Application Security | OWASP Top 10 |
| Indian Cybersecurity Guidelines | CERT-In Security Guidelines |

| | |
|---|---|
| Information Security Standard | ISO/IEC 27001:2022 |
| **Data Privacy Regulation** | **Digital Personal Data Protection (DPDP) Act, 2023** |
| Healthcare Security Standard | HIPAA Security Rule |
| Payment Security Standard | PCI-DSS v4.0 |

## Software & Tools Used

| Category | Tool |
|---|---|
| Cloud Platforms | Microsoft Azure, Amazon Web Services (AWS) |
| Infrastructure as Code | Terraform |
| Programming Language | Python |
| Machine Learning | Scikit-learn (Random Forest) |
| Data Processing | Pandas, NumPy |
| CSPM Tools | Prowler, Steampipe, ScoutSuite |
| **AI/LLM** | **OpenAI GPT,Perplexity,Gemini,Claude** |
| Automation | Azure Functions, AWS Lambda |
| Version Control | Git, GitHub |
| Development Environment | Visual Studio Code |

## References


- NIST Cybersecurity Framework 2.0
- CIS Benchmarks
- MITRE ATT&CK
- OWASP Top 10
- ISO/IEC 27001:2022
- DPDP Act 2023